

Waste Genie: A Web-Based Educational Technology for Sustainable Waste Management

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ABSTRACT

In this work, we proposed and designed a web-based educational technology to support sustainable living. It is called Waste Genie. Waste Genie integrates formal and informal learning features to help users learn about waste management. A field trial was conducted to evaluate its effectiveness, and the results indicated the platform was well-received by the users.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in collaborative and social computing.**

KEYWORDS

educational technology, sustainability learning, interactive feedback, waste management.

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1 INTRODUCTION

Waste management is a key factor to practice sustainable living. However, it is extremely challenging. For instance, waste classifying can be confusing to identify the correct categories, due to many reasons that not all plastic is recyclable, and recycling regulations vary across the counties. Researchers have utilized emerging technologies to optimize the waste management process, including deep learning models for plastic wastes [13], COVID wastes [4], and recyclables [17], etc. Some applications were also proposed to educate people with waste management knowledge, such as identifying recycling options for items selling in vending machines[5], AR game for recycling education[18], and smart trash bin to increase people's awareness toward environment[16]. In addition, as the development of mobile computing, social media, immersive technologies, and educational technologies, the boundary between formal and informal learning has been blurred [10]. With the help of web technologies [7] and social media features [6, 8, 9], students

demonstrated better engagement and motivation from the informal learning process. All of the above mentioned background, motivates us to research a technological solution that marries both formal and informal learning features to help people learn about waste management.

In this work, we designed Waste Genie, a web-based educational technology to support environmental sustainability learning. It offers formal instructional content and informal social elements in the technology. We examined the impacts of static and interactive information on students and usability of the application through data analysis.

2 METHODOLOGY

2.1 The Waste Genie Application

Our proposed web application - Waste Genie, is designed to help people learn waste management and practice waste sorting. It was developed using Flutter and then compiled as a web-app.

Fig. 1 demonstrates the key features in the app. The homepage (Fig. 1a) integrates a feed containing static posts and interactive quizzes. The posts are organized in a chronological order. For each post, users can like them by pressing the heart button and comment below it to start a discussion. Through our dedicated editor (Fig. 1b & 1c), users can create posts related to sustainable living, or create quizzes to challenge others on waste sorting. To complete a quiz, users can drag and drop the category label to the waste objects (Fig. 1e & 1f). Through the completion of the quizzes, the virtual "carbon credits" will be awarded. Such credits are estimated based on EPA's data on the amount of green gas benefits generated from recycling, landfilling, and composting [2]. It roughly demonstrates how much green house gas (in gCO₂e, grams of carbon dioxide equivalent) can be reduced if we process the garbage properly. Each user can find their ranking on the carbon credits they earned in the leaderboard (Fig. 1d). The credits are also converted to tree-days and miles in car by applying the green algorithm [12], to provide a more intuitive illustration of the carbon benefits. Users can also use the app to look up for the waste-bin association of a specific waste item, by either directly search it (Fig. 1g) or use the built-in waste scanner to take a photo (Fig. 1h). The scanner was trained under EfficientNet [15] using several open-source waste dataset [1, 11, 14]. It automatically detects the possible waste types and gives suggestions on how to properly discard them.

2.2 User Study

We recruited 28 students from a web usability class to try Waste Genie for 13 days and recorded their activities on the app. During their usage, each student is asked to create a static post and an interactive quiz. They were also encouraged to use any features

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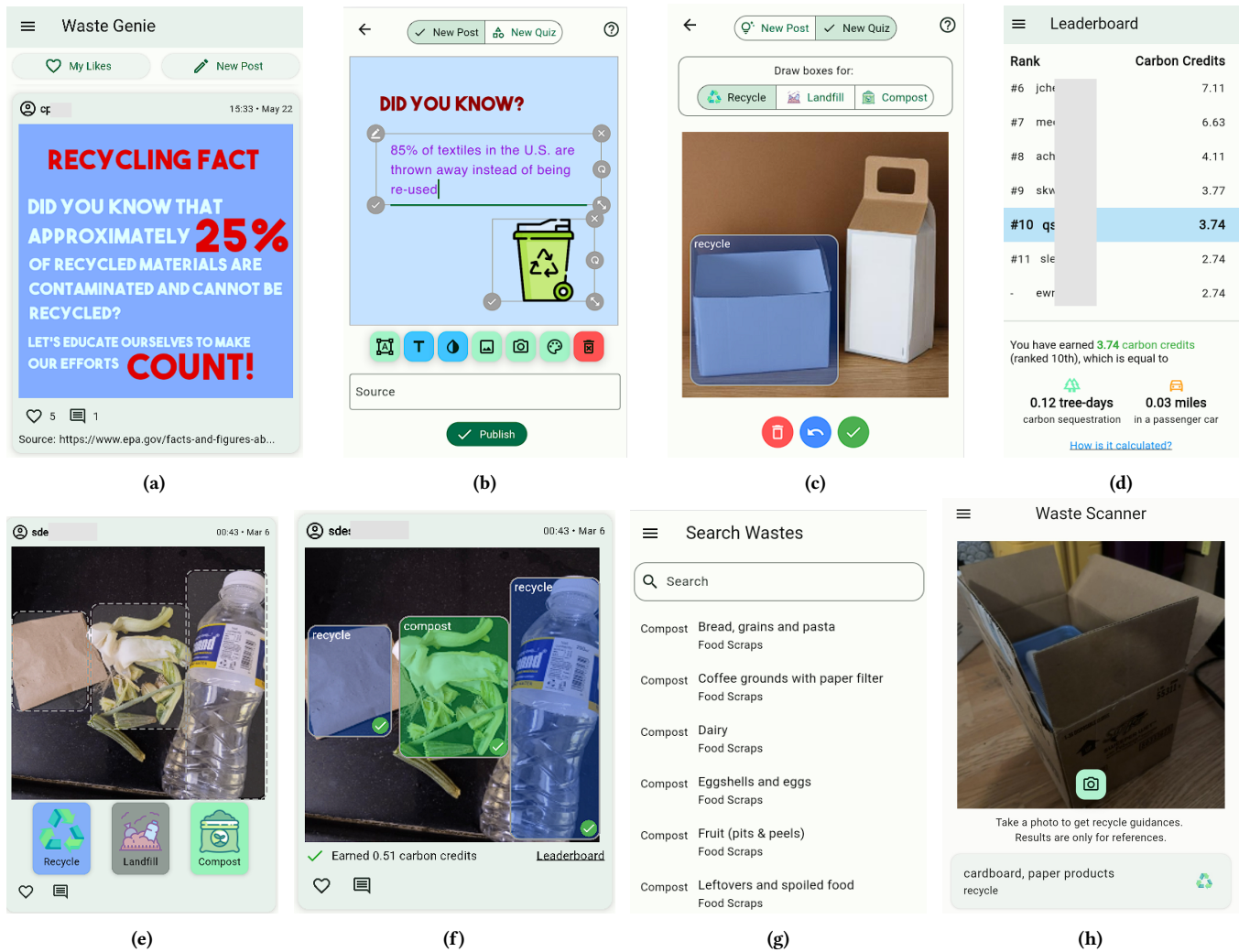


Figure 1: Waste Genie web-app: (a) The homepage with a feed; (b & c) The editor for static post and interactive quiz; (d) The leaderboard of carbon credits; (e & f) Users can drag and drop the labels onto the items to practice waste sorting skills; (g) Directly search for wastes and their correct classifications; (h) The waste scanner.

they would like to. After the 13-day study, a survey containing the System Usability Scale questions [3] was also distributed to the students to collect their feedback. The questions were asked in a 5-point Likert scale, where 1 means strongly disagree, and 5 means strongly agree. This post-survey aimed to access the usability of the app, overall user experience, students' satisfaction, and collect suggestions from the students.

3 EVALUATIONS & RESULTS

During the 13-day experiment, each user spent 8.8 minutes on average viewing the feed and 6.2 minutes creating the contents. Among the usage time of all the features, 59.7% of them are consuming contents (reading the feed, taking the quiz, etc.).

In the survey distributed to the students after the study, most students held a positive attitude toward the application. Users found

Waste Genie easy to use ($M = 4.4, SD = 0.63$), and expressed confidence while using it ($M = 4.28, SD = 0.72$). Students indicated that they did not require the support of a technical person ($M = 1.4, SD = 0.63$), and they can expect most people would learn to use the app very quickly ($M = 4.4, SD = 0.63$). Participants also rated the integration and consistency of various features in the application highly, with a mean score of 4.12 ($SD = 0.82$) for integration and 1.68 ($SD = 0.79$) for the feelings of inconsistency.

Overall, users demonstrated a good impression on Waste Genie. Several points were also suggested by the students for further improvements. These include enhancing the app's visual elements, improving the performance on web browsers, providing clearer instructions and guidance, increasing the accuracy of the waste scanner, and integrating with popular apps. By addressing these suggestions, we will improve Waste Genie's overall user experience and engagement in the future.

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